

APPROVAL WORKFLOWS & GOVERNANCE

Module 11 | Functional boards, workflows and governance controls

A consolidated reference for product, business, UX, architecture, development, testing and operational stakeholders.

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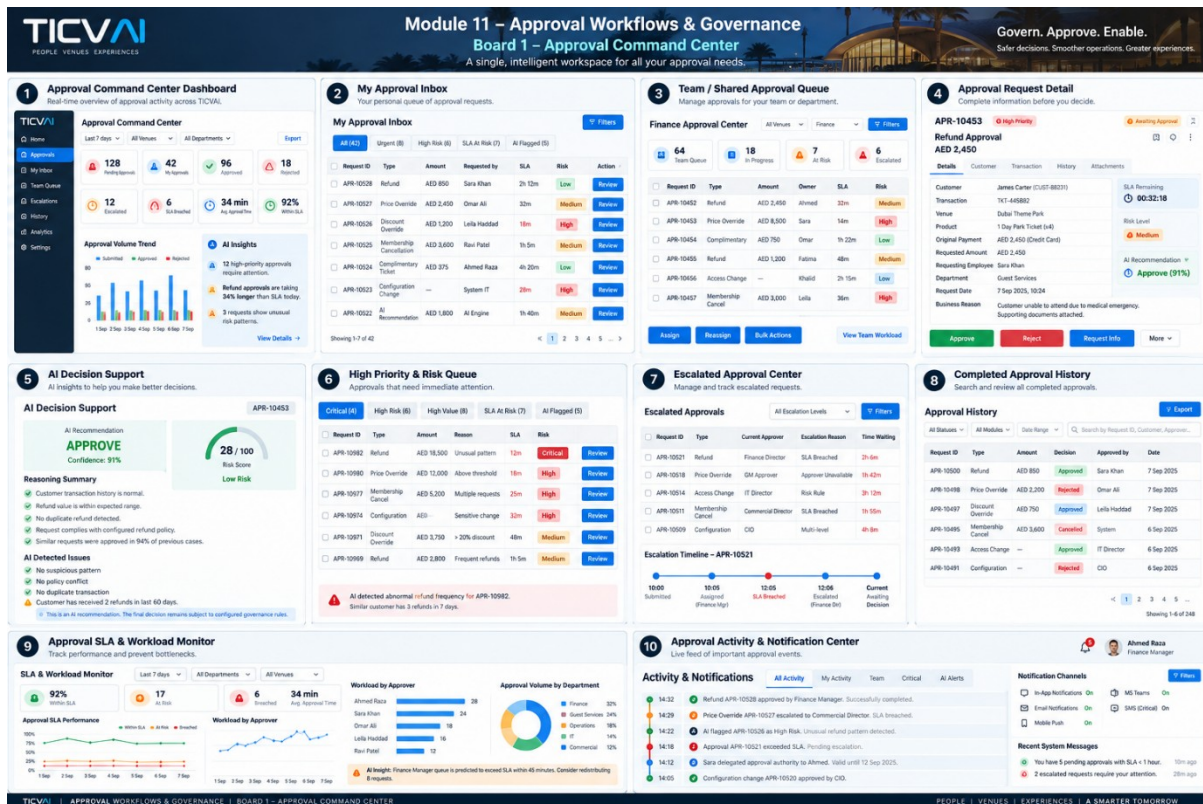
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Module 11 — Approval Workflows & Governance

Board 1 — Approval Command Center



Board Objective

Provide a centralized workspace where authorized users can **view, prioritize, review, action, monitor, and track approval requests across the TICVAI platform.**

The Command Center shall consolidate approvals originating from Ticketing, Finance, POS, Membership, Access Control, Pricing, Promotions, Configuration, AI recommendations, and other TICVAI modules into a single governed approval environment.

The matrix specifically requires dashboards covering pending, approved, rejected and escalated requests, together with notifications, reminders and outcome notifications.

Screen 1 — Approval Command Center Dashboard

Purpose

Provide management and approvers with a real-time overview of approval activity.

Key Functions

- Pending approvals
- Awaiting my approval
- Approved today
- Rejected
- Escalated
- SLA breached
- High-risk requests
- AI-prioritized requests
- Average approval time
- Approval volume
- Approval trend
- **Filter by:**
 - o Tenant
 - o Venue
 - o Department
 - o Module
 - o Request type
 - o Priority
 - o Risk
 - o **Status**
 - o Date

AI Insight Panel

Examples:

12 high-priority approvals require attention.

Refund approvals are taking 34% longer than the configured SLA today.

3 requests show unusual risk patterns.

Screen 2 — My Approval Inbox

Purpose

Provide each approver with their personal actionable queue.

Key Functions

Display:

- Request ID
- Request type
- Requested by
- Venue

- Department
- Requested value
- Submitted time
- SLA remaining
- Risk level
- Priority
- Current approval stage

Tabs:

All | Urgent | High Risk | SLA At Risk | AI Flagged

Actions:

Review | Approve | Reject | Delegate | Escalate

The system shall notify approvers when new requests are assigned and provide reminders for pending approvals.

Screen 3 — Team / Shared Approval Queue

Purpose

Allow managers and centralized approval teams to manage approvals belonging to their team or department.

Example:

Finance Approval Center

Request	Type	Amount	Owner	SLA	Risk
APR-1045 2	Refund	AED 2,450	Ahmed	32 min	Medium
APR-1045 3	Price Override	AED 8,500	Sara	14 min	High
APR-1045 4	Complimentary	AED 750	Omar	1h 22m	Low

Functions

- Reassign request
- Claim request
- Bulk assignment
- Team workload
- Approver availability
- Queue aging
- SLA monitoring

This also prepares the architecture for the matrix requirement covering **Shared Service Approval Centers**.

Screen 4 — Approval Request Detail

This should be one of the most important screens on the board.

Header

APR-10453 — Refund Approval

AED 2,450

Status: **Awaiting Approval**

Risk: **Medium**

SLA Remaining: **00:32:18**

Request Information

- Customer
- Transaction
- Ticket/order
- Venue
- Product
- Original payment
- Requested amount
- Requesting employee
- Department
- Request date/time
- Business reason

Supporting Information

- Attachments
- Previous transactions
- Related requests
- Customer history
- Previous approval history

Decision Area

Approve

Reject

Request Information

Escalate

Delegate

A rejection should require the approver to provide a reason, as required by the source matrix.

Screen 5 — AI Decision Support

This should distinguish TICVAI from a conventional workflow product.

The AI should **assist the approver rather than automatically replace governance controls**.

Example

AI Recommendation

APPROVE

Confidence: 91%

Reasoning Summary

- Customer transaction history is normal.
- Refund value is within expected range.
- No duplicate refund detected.
- Request complies with configured refund policy.
- Similar requests were approved in 94% of previous cases.

Risk Score

28 / 100 — LOW

AI Detected Issues

- ✓ No suspicious pattern
- ✓ No policy conflict
- ✓ No duplicate transaction
- △ Customer has received 2 refunds during previous 60 days

The requirements explicitly call for AI-generated approval recommendations, AI risk assessment and AI priority scoring.

Important: the final decision remains governed by the configured approval workflow.

Screen 6 — High Priority & Risk Queue

Purpose

Create a dedicated operational screen for approvals requiring immediate attention.

Categories:

Critical

High Risk

High Value

SLA At Risk

Escalated

AI Flagged

Example

APR-10982 — Refund AED 18,500

High Risk

SLA expires in 12 minutes

AI detected abnormal refund frequency.

Management can sort requests by:

Risk × Financial Value × SLA × Business Impact

This complements the matrix's value-based, risk-based and conditional routing requirements.

Screen 7 — Escalated Approval Center

Purpose

Manage requests that were escalated because of SLA breaches, risk conditions, unavailable approvers or configured escalation rules.

Display:

- Original approver
- Current approver
- Escalation level
- Escalation reason
- Time waiting
- SLA status
- Previous actions
- Next escalation level

Escalation Timeline

10:00 — Request Submitted

↓

10:05 — Finance Manager Assigned

↓

12:05 — SLA Breached

↓

12:06 — Escalated to Finance Director

↓

Current — Awaiting Decision

The matrix requires multi-level escalation together with complete escalation history, reporting and analytics.

Screen 8 — Completed Approval History

Purpose

Provide searchable historical records of completed approval decisions.

Statuses:

Approved | Rejected | Cancelled | Withdrawn | Expired

Search By

- Request ID
- Transaction
- Customer
- Requester
- Approver
- Venue
- Department

- Module
- Date
- Decision
- Approval amount

Each record should preserve:

Requester → Workflow → Approvers → Comments → Decision → Timestamp → Executed Action

The matrix requires a complete audit trail containing requests, approvers, comments, timestamps and decisions.

Screen 9 — Approval SLA & Workload Monitor

Purpose

Give managers visibility into operational performance **before approvals become bottlenecks**.

KPI Cards

Within SLA — 92%

At Risk — 17

Breached — 6

Average Approval — 34 min

Operational Views

Approval workload by:

- Approver
- Team
- Department
- Venue
- Approval type

Example AI insight:

Finance Manager queue is predicted to exceed SLA within 45 minutes. Consider redistributing 8 requests.

This supports configurable approval response-time targets and breach alerts.

Screen 10 — Approval Activity & Notification Center

Purpose

Provide a unified chronological feed of important approval events.

Example:

14:32

Refund APR-10528 approved by Finance Manager.

14:29

Price Override APR-10527 escalated to Commercial Director.

14:22

AI flagged APR-10526 as High Risk.

14:18

Approval APR-10521 exceeded SLA.

14:12

Sara delegated approval authority to Ahmed until 12 September.

Filters**My Activity | Team | Venue | Department | Critical | AI Alerts**

Notifications can be delivered through the appropriate configured channels, while the detailed channel/integration configuration can be handled in later boards.

Board 1 — Screen Summary

#	Screen
1	Approval Command Center Dashboard
2	My Approval Inbox
3	Team / Shared Approval Queue
4	Approval Request Detail
5	AI Decision Support
6	High Priority & Risk Queue
7	Escalated Approval Center
8	Completed Approval History
9	Approval SLA & Workload Monitor
10	Approval Activity & Notification Center

Acceptance Criteria

Board 1 shall be considered complete when authorized users can centrally access approval requests from multiple TICVAI modules; distinguish personal, team, high-risk and escalated queues; inspect the complete business context before making a decision; approve/reject/escalate/delegate according to permissions; view AI recommendations and risk indicators without bypassing configured governance; monitor SLA and workload; and access a traceable history of approval actions.

Boundary with the next board: Board 1 is primarily “**operate the approvals.**” We should avoid putting workflow-building configuration here. **Board 2 will be the Approval Workflow Builder**, where administrators configure triggers, approval levels, sequential/parallel paths, conditions, actions, and workflow versions.

Module 11 — Approval Workflows & Governance

Board 2 — Approval Workflow Builder

Module 11 – Approval Workflows & Governance
Board 2 – Approval Workflow Builder
 Configure. Automate. Govern.

1 Approval Workflow Library
 Manage all approval workflows across your organization.

2 Create Approval Workflow
 Step-by-step wizard to define a new approval workflow.

3 Visual Workflow Designer
 Build approval workflows using a drag-and-drop interface.

4 Approval Stage Configuration
 Define approver details and stage settings.

5 Condition & Decision Rule Builder
 Set conditions to control the approval routing.

6 Approval Sequence & Parallel Routing
 Configure sequential, parallel and multi-level approvals.

7 Workflow Outcome & Action Configuration
 Define actions to be executed after approval decisions.

8 Workflow Validation & Simulation
 Test the workflow with test scenarios before activation.

9 Workflow Publication & Lifecycle
 Manage the workflow lifecycle and deployment.

10 Workflow Versioning & Change History
 Track changes and maintain full audit history.

Board Objective

Board 2 shall provide administrators with a **central visual configuration environment** to create, modify, test, approve, publish, and version approval workflows used throughout TICVAI.

While Board 1 is the operational workspace for processing approvals, **Board 2 defines how those approvals work.**

The platform shall support configurable workflows, including both single-level and multi-level approval chains. The key principle is that administrators should configure workflows visually rather than requiring development or hard-coded rules.

Screen 1 — Approval Workflow Library

Purpose

Provide a centralized directory of every approval workflow configured within the tenant.

Information Displayed

- Workflow Name
- Workflow ID

- Business Process
- Source Module
- Tenant
- Venue / Department
- Current Version
- **Status**
- Effective Date
- Last Modified
- Owner

Example Workflows

Workflow	Module	Status	Version
Refund Approval	Finance	Active	V4
Price Override	Ticketing	Active	V2
Discount Override	Promotions	Active	V3
Complimentary Ticket	Ticketing	Active	V1
Access Permission Change	Access Control	Active	V2
AI Pricing Recommendation	Dynamic Pricing	Draft	V3

Actions

Create Workflow | Clone | Edit | Disable | Archive | View History

Filters shall support module, venue, department, tenant, status and workflow type.

Screen 2 — Create Approval Workflow

Purpose

Provide a guided setup wizard for creating a new approval workflow.

Step 1 — Basic Information

Configure:

- Workflow Name
- Description
- Business Process
- Source Module
- Request Type
- Tenant
- Department
- Venue
- Workflow Owner

Step 2 — Applicability

Define where the workflow applies.

Example:

Business Process: Refund

Channel: POS + B2C

Venue: All Venues

Effective From: 01 October 2026

Step 3 — Governance

Configure:

- Require approval
- Allow automatic approval
- Allow delegation
- Allow escalation
- Require comments
- Require rejection reason
- Require MFA
- Require digital signature

The matrix requires authentication, MFA and sensitive-action confirmation for approval actions.

Screen 3 — Visual Workflow Designer**Purpose**

This should be the **main configuration screen of Board 2**.

Administrators shall build approval processes using a drag-and-drop workflow canvas.

Component Palette

Trigger

Condition

Approval

Parallel Approval

Decision

Escalation

Notification

Execute Action

End

Example

Refund Requested

↓

Check Refund Amount

↓

≤ AED 500



The visual builder should immediately show invalid workflow paths or missing approval stages.

Screen 4 — Approval Stage Configuration

Purpose

Configure each approval stage in detail.

When the administrator clicks an **Approval Node** in Screen 3, this screen/panel opens.

Configuration

Stage Name: Finance Manager Approval

Approver Selection Method:

- Specific user
- Role
- Department
- Organizational hierarchy
- Manager of requester
- Approval group
- Shared approval center

Approval Mode

Single Approver

Any One Approver

All Approvers

Majority

Minimum X Approvers

Example

Required Approvers: Finance Manager + Venue Manager

Completion Rule: Both must approve.

The source specifically supports sequential, parallel, consensus and majority-based approval models.

Screen 5 — Condition & Decision Rule Builder

Purpose

Allow administrators to determine **when a particular approval path should apply**.

The rule builder should use business-friendly configuration rather than programming.

Example

IF

Refund Amount

IS GREATER THAN

AED 5,000

AND

Risk Score

IS GREATER THAN

70

THEN

Require Finance Director Approval.

Available Conditions

- Transaction amount
- Discount percentage
- Refund percentage
- Product
- Ticket type
- Venue
- Department
- Channel
- Customer type
- Membership level
- Requesting user
- User role
- Risk score
- AI risk score
- Date/time
- Business rule result

The matrix explicitly requires value-based, risk-based and conditional approval rules.

Screen 6 — Approval Sequence & Parallel Routing

Purpose

Configure complex multi-level approval structures.

Administrators can define:

Sequential

Supervisor

↓

Finance Manager

↓

Finance Director

↓

GM

Parallel

After Finance Manager:

✓ **Finance Director**

↘ **Operations Director**

Both can review simultaneously.

Consensus

All configured approvers must approve.

Majority

Example:

3 of 5 approvers required

The workflow shall visually display the complete routing structure and completion criteria.

This directly supports the approval execution models defined in the requirements.

Screen 7 — Workflow Outcome & Action Configuration

Purpose

Define what TICVAI should actually do after the approval process reaches an outcome.

This is important because **approval and execution should be separated**.

APPROVED

Possible actions:

- Execute refund
- Apply price override
- Apply discount
- Issue complimentary ticket
- Cancel membership
- Modify access permission

- Publish configuration
- Execute AI recommendation
- Call API
- Publish event
- Send notification

REJECTED

Possible actions:

- Cancel requested action
- Return request to requester
- Allow modification/resubmission
- Notify requester
- Record rejection reason

EXPIRED

Possible actions:

- Cancel request
- Escalate request
- Restart workflow
- Notify owner

The requirements include withdrawal, resubmission, cancellation, expiration and reopening of approval requests.

Screen 8 — Workflow Validation & Simulation

Purpose

Allow administrators to test workflows **before putting them into production**.

This is something I strongly recommend adding to the UX because these workflows can directly affect money, access and customer operations.

Test Scenario

Action: Refund

Amount: AED 7,500

Venue: Dubai Theme Park

Requester: POS Supervisor

Risk Score: 42

Simulation Result

Step 1

Refund workflow triggered ✓

Step 2

Amount > AED 5,000 ✓

Step 3

Finance Manager required ✓

Step 4

Finance Director required ✓

Step 5

MFA required ✓

Expected Route

POS Supervisor → Finance Manager → Finance Director → Execute Refund

Validation Checks

- Missing approver
- Circular workflow
- Dead-end path
- Invalid condition
- Conflicting rules
- Missing escalation route
- Missing outcome
- SoD violation

This screen is an important design recommendation even though the matrix does not explicitly name a simulation screen.

Screen 9 — Workflow Publication & Lifecycle

Purpose

Govern the transition of workflow configurations into production.

Lifecycle**Draft**

→ Under Review

→ Approved

→ Scheduled

→ Active

→ Suspended

→ Retired

Configuration

- Effective From
- Effective Until
- Tenant
- Venue
- Deployment scope
- Activation schedule

Publication Summary

Before activation:

Refund Approval V5

Changes:

- Finance Director threshold changed from AED 5,000 → AED 7,500.
- MFA added for refunds > AED 10,000.
- SLA changed from 2 hours → 1 hour.

Affected Venues: 8

Affected Users: 42

Actions:

Submit for Approval | Schedule | Publish | Cancel

This also allows the workflow configuration itself to be governed by TICVAI's approval framework.

Screen 10 — Workflow Versioning & Change History

Purpose

Maintain complete governance over changes made to approval workflows.

The matrix specifically requires **Approval Workflow Versioning**.

Version History

Version	Status	Changed By	Date
V5	Active	Admin	7 Sep 2026
V4	Retired	Sara	18 Aug 2026
V3	Retired	Ahmed	2 Jul 2026

Version Comparison

V4 → V5

Threshold:

~~AED 5,000~~

AED 7,500

MFA:

~~Not Required~~

Required > AED 10,000

SLA:

~~2 Hours~~

1 Hour

Actions

Compare | View | Clone | Restore / Rollback

The audit record should capture:

Who changed it → What changed → When → Why → Approval → Publication

Board 2 — Screen Summary

#	Screen
1	Approval Workflow Library
2	Create Approval Workflow
3	Visual Workflow Designer
4	Approval Stage Configuration
5	Condition & Decision Rule Builder
6	Approval Sequence & Parallel Routing
7	Workflow Outcome & Action Configuration
8	Workflow Validation & Simulation
9	Workflow Publication & Lifecycle
10	Workflow Versioning & Change History

Board 2 Acceptance Criteria

Board 2 shall be considered complete when an authorized administrator can create an approval workflow **without development intervention**, define its business context and trigger, visually construct approval paths, configure single/multi-level and sequential/parallel approval stages, establish conditional routing, define post-approval actions, simulate the workflow before activation, govern publication, and maintain full version history.

Important Board Boundary

We should **not overload Board 2 with the complete organizational approval matrix**.

Board 2 answers:

“How does this workflow operate?”

Board 3 — Approval Matrix & Intelligent Routing will answer:

“Who has authority to approve what, at which tenant, venue, department, monetary threshold, risk level and organizational level?”

That separation will prevent duplication and keep the configuration architecture clean.

Module 11 — Approval Workflows & Governance

Board 3 — Approval Matrix & Intelligent Routing

The screenshot displays the TICVAI Module 11 - Approval Workflows & Governance interface, specifically Board 3 - Approval Matrix & Intelligent Routing. The interface is divided into ten numbered sections, each representing a different aspect of the approval workflow configuration:

- 1. Approval Matrix Command Center:** Provides a high-level overview of the approval matrix, including active matrices, rules, and requests.
- 2. Approval Authority Matrix:** Allows users to define who can approve each action and up to which limit, with a table showing approval levels and thresholds.
- 3. Organizational Hierarchy Routing:** Visualizes the organizational hierarchy and routing paths for approvals.
- 4. Department-Based Approval Matrix:** Configures approval rules based on department, with a table showing rules and their status.
- 5. Venue & Tenant Approval Matrix:** Manages approval policies by tenant and venue with inheritance, including a table for venue rules.
- 6. Value & Threshold Routing:** Configures monetary or percentage thresholds for routing, with a table showing value ranges and required approvals.
- 7. Risk-Based & Conditional Routing:** Adds conditions based on risk, customer, product or other factors, with a table for risk rules.
- 8. Approver Group & Decision Policy:** Defines multi-approver groups and decision rules, including a table for group members and decision policies.
- 9. Routing Simulator & Conflict Detection:** Tests the approval matrix with real scenarios, showing a test scenario and routing result.
- 10. AI Routing Advisor & Matrix Optimization:** Provides insights to improve approval matrices, including a table for AI insights and optimization opportunities.

Board Objective

Board 3 shall provide administrators with a centralized environment to define **who is authorized to approve each type of request, under which business conditions, and at which organizational level.**

Board 2 defines **how the workflow operates.** Board 3 defines **who has approval authority and how TICVAI dynamically identifies the correct approver.**

The source specifically requires configurable approval matrices based on organizational structure, hierarchy, department, venue and tenant.

The board shall also support value-based, risk-based and conditional routing.

Screen 1 — Approval Matrix Command Center

Purpose

Provide administrators with a centralized overview of all approval matrices configured across TICVAI.

KPI Cards

- Active Approval Matrices
- Approval Rules

- Tenants Covered
- Venues Covered
- Departments Covered
- Unassigned Rules
- Routing Conflicts
- Governance Warnings

Matrix Directory

Display:

Matrix	Process	Scope	Rules	Status
Refund Authority	Refund	All Venues	12	Active
Discount Authority	Promotions	Retail	8	Active
Price Override	Ticketing	Dubai Park	6	Active
Access Changes	Access Control	All Venues	5	Active
AI Recommendations	AI Engine	Corporate	9	Active

Actions

[Create Matrix](#) | [Edit](#) | [Clone](#) | [Validate](#) | [Activate](#) | [Archive](#)

Screen 2 — Approval Authority Matrix

Purpose

Provide a visual table showing **who can approve what and up to which authority level**.

Example:

Action	Supervisor	Manager	Director	GM
Refund	AED 500	AED 5K	AED 20K	Unlimited
Discount	10%	20%	35%	>35%
Price Override	5%	10%	20%	>20%
Complimentary	AED 250	AED 2K	AED 10K	Unlimited

Administrators can click any cell to modify its authority rule.

Configuration Dimensions

- Role
- User
- Department
- Venue

- Tenant
- Business process
- Transaction value
- Percentage
- Risk level

This becomes the client's **central Delegation of Authority / Approval Matrix**.

Screen 3 — Organizational Hierarchy Routing

Purpose

Configure approval routing based on the client's organizational hierarchy.

Example:

POS Cashier

↓

POS Supervisor

↓

Venue Operations Manager

↓

Operations Director

↓

General Manager

Administrators shall be able to configure:

- Reporting hierarchy
- Manager relationships
- Approval levels
- Substitute roles
- Maximum authority
- Hierarchy inheritance

Dynamic Routing Example

Requester:

Sara — POS Supervisor

TICVAI identifies:

Manager → Ahmed

Director → Fatima

GM → Omar

The system then routes the request automatically.

The matrix explicitly requires routing according to organizational hierarchy.

Screen 4 — Department-Based Approval Matrix

Purpose

Allow different departments to maintain different approval authority structures.

Department Tabs

Finance | Operations | Commercial | Marketing | IT | Security | Guest Services

Example:

Finance

Refund > AED 5,000

→ Finance Director

Commercial

Discount > 20%

→ Commercial Director

IT

Critical configuration change

→ IT Director

Security

Access permission modification

→ Security Manager

This is important because the same financial threshold or approval level should not necessarily apply to every department.

Department-specific workflows are explicitly required.

Screen 5 — Venue & Tenant Approval Matrix

Purpose

Allow approval policies to differ between TICVAI clients and between venues belonging to the same client.

Hierarchy

Tenant

→ **Business Unit**

→ **Venue**

→ **Department**

Example:

Tenant: Dubai Leisure Group

Global Policy

↓

Dubai Theme Park

Venue Override

↓

Waterpark

Venue Override

↓

Museum

Uses Global Policy

Inheritance

Administrators should see:

Inherited**Overridden****Local****Not Configured****Example**

Corporate refund threshold:

AED 5,000

Waterpark override:

AED 3,000

Museum:

Inherited AED 5,000

The matrix requires both venue-specific and tenant-specific approval workflows.

This screen must respect strict multi-tenant isolation.

Screen 6 — Value & Threshold Routing

Purpose

Configure approval authority according to transaction value or percentage thresholds.

Example — Refund

Amount	Required Approval
AED 0–500	Supervisor
AED 501–2,500	Manager
AED 2,501–10,000	Finance Director
AED 10,001–25,000	CFO
> AED 25,000	CFO + GM

Example — Discount**0–10% → Automatic / Supervisor****11–20% → Commercial Manager****21–30% → Commercial Director****>30% → GM**

The administrator should be able to drag threshold boundaries or enter exact values.

Validation

TICVAI should detect:

- Threshold gaps
- Overlapping thresholds
- Missing approvers
- Unlimited authority
- Conflicting rules

Value-based approval routing is explicitly required by the matrix.

Screen 7 — Risk-Based & Conditional Routing

Purpose

Dynamically modify approval routing based on the risk or context of the request.

Example

Normal refund:

AED 1,500 → Finance Manager

But:

AED 1,500 + High Risk

↓

Finance Manager + Finance Director

Rule Builder

IF

Risk Score > **80**

OR

Customer Refund Frequency > **3**

OR

Transaction flagged for fraud

THEN

Add **Finance Director**

AND

Require **MFA**

AND

Disable automatic approval.

Other Conditions

- Customer type
- Channel
- Product
- Event
- Venue
- Transaction history
- User role

- Request frequency
- Time/date
- AI risk score

Risk-based routing is directly required by the source.

Screen 8 — Approver Group & Decision Policy

Purpose

Define cases where approval authority belongs to multiple people rather than one individual.

Approval Group

Executive Refund Committee

Members:

- Finance Director
- Operations Director
- Commercial Director
- General Manager
- CFO

Decision Policy

Administrators select:

Any One

All / Consensus

Majority

Minimum X of Y

Specific Mandatory Approver + Majority

Example:

Refund > AED 50,000

Required:

CFO — Mandatory

-

2 of 3 Directors

The matrix explicitly supports parallel, consensus and majority approval models.

Screen 9 — Routing Simulator & Conflict Detection

Purpose

Allow administrators to test the complete approval matrix before deployment.

This is particularly important because Board 3 can contain hundreds of combinations.

Test Scenario

Tenant: Dubai Leisure Group

Venue: Waterpark

Department: Finance
Action: Refund
Amount: AED 12,500
Requester: POS Supervisor
Risk: High
AI Risk Score: 82

TICVAI Routing Result

Step 1

Venue rule identified ✓

Step 2

Amount > AED 10,000 ✓

Step 3

High-risk rule triggered ✓

Step 4

Finance Director required ✓

Step 5

CFO required ✓

Step 6

MFA required ✓

Final Route

Requester

↓

Finance Director

↓

CFO

↓

Execute Refund

Conflict Detection

The system shall identify:

- No eligible approver
- Conflicting authority rules
- Overlapping thresholds
- Circular routing
- Missing hierarchy
- Self-approval risk
- Tenant/venue override conflict

Screen 10 — AI Routing Advisor & Matrix Optimization

Purpose

Use TICVAI AI to analyze approval behavior and recommend improvements to the approval matrix.

The source requires AI risk assessment, priority scoring and escalation recommendations.

AI Insights

Example:

Potential Bottleneck

Finance Director currently receives 42% of all refund approvals.

Recommendation

Increase Finance Manager refund authority from AED 5,000 to AED 7,500.

Expected Impact

Approvals routed to Director: ↓ 23%

Average approval time: ↓ 18 min

Risk impact: Low

Another example:

Governance Risk Detected

Two roles currently have unlimited complimentary-ticket approval authority.

Recommended Action

Introduce an AED 10,000 threshold and require dual approval above this value.

Important Governance Principle

AI should **recommend**, not silently modify the authority matrix.

Actions:

Review Recommendation

Simulate Impact

Accept & Create Draft Change

Reject

Any accepted AI recommendation can itself be sent through the governance approval workflow before publication.

Board 3 — Screen Summary

#	Screen
1	Approval Matrix Command Center
2	Approval Authority Matrix
3	Organizational Hierarchy Routing
4	Department-Based Approval Matrix
5	Venue & Tenant Approval Matrix
6	Value & Threshold Routing
7	Risk-Based & Conditional Routing
8	Approver Group & Decision Policy

#	Screen
9	Routing Simulator & Conflict Detection
10	AI Routing Advisor & Matrix Optimization

Board 3 Acceptance Criteria

Board 3 shall be considered complete when authorized administrators can define and maintain approval authority based on **tenant, venue, department, organizational hierarchy, role, user, transaction value, percentage, business conditions and risk**; configure multi-person approval policies; inherit or override corporate policies at venue level; test the resulting routing before activation; and identify missing, conflicting or unsafe approval rules.

The platform shall automatically determine the appropriate approver or approval group at runtime while maintaining tenant isolation. The matrix specifically requires approval isolation between tenants.

Board Boundary

This separation remains important:

Board 2 — Workflow Builder:

“What steps does the approval process follow?”

Board 3 — Approval Matrix & Intelligent Routing:

“Who is authorized to approve, and under what conditions?”

Board 4 should move to **Approval Execution & Decision Management**: the actual decision experience, evidence, comments, rejection reasons, withdrawal, resubmission, and controlled execution of the approved business action.

Module 11 — Approval Workflows & Governance

Board 4 — Approval Execution & Decision Management

Board Objective

Board 4 shall manage the **actual approval decision lifecycle** after a request reaches an approver.

Boards 2 and 3 determine the workflow and routing. Board 4 answers:

“Now that the request has reached me, what information do I need, what decision can I make, what happens after my decision, and how is everything recorded?”

The matrix requires comments, mandatory rejection reasons, withdrawal, resubmission, segregation of duties, Four-Eyes control, draft/cancel/expire/reopen functions, authentication, MFA, and sensitive-action confirmation.

I recommend the following **10 screens**.

Screen 1 — Approval Decision Workspace

Purpose

Provide the approver with one comprehensive workspace to evaluate and action a request.

Header

APR-10543 — Refund Approval

AED 7,500

Status: **Awaiting My Approval**

Priority: **High**

Risk: **Medium**

SLA Remaining: **00:42:16**

Main Sections

Request Summary

- Request type
- Transaction/order
- Customer
- Venue
- Department
- Requester
- Amount/value
- Request date
- Business reason

Approval Context

- Current approval stage
- Previous approvers
- Remaining approvers
- Applicable workflow
- Authority rule
- SLA

Primary Actions

Approve | Reject | Request Information | Escalate | Delegate

The important design principle is that an approver should **not need to leave this screen** to understand the request.

Screen 2 — Business Context & Evidence Viewer

Purpose

Provide the business evidence required to make an informed decision.

For example, for a refund approval:

Transaction Information

- Original order
- Ticket/product
- Purchase channel
- Payment method
- Original amount
- Used/unused status
- Visit date

- Previous modifications

Customer Information

- Customer profile
- Membership
- Transaction history
- Previous refunds
- Previous disputes

Supporting Evidence

- Documents
- Images
- Emails
- Notes
- Attachments
- Supporting references

Related Activity

TICVAI should automatically display related requests and previous approvals where relevant.

Screen 3 — Approval Timeline & Decision Chain

Purpose

Provide a visual representation of the complete approval journey.

Example:

Sara — Request Submitted

10:05 ✓

↓

Finance Manager — Approved

10:22 ✓

↓

Finance Director — Awaiting Approval

Current

↓

General Manager — Required if > AED 10K

↓

Execute Refund

Timeline Events

Display:

- Request created
- Workflow triggered
- Routing decision
- Assignment
- Reassignment

- Delegation
- Comments
- Approval
- Rejection
- Escalation
- Notification
- Execution

The underlying matrix requires a complete history containing approvers, comments, timestamps and decisions.

Screen 4 — Approve & Sensitive Action Confirmation

Purpose

Control the final approval action before it becomes binding.

When the approver selects:

APPROVE

TICVAI displays a confirmation panel.

You are approving:

Refund: AED 7,500

Customer: James Carter

Venue: Dubai Theme Park

Request: APR-10543

Confirmation Requirements

Depending on policy:

- Approval comment
- Password confirmation
- MFA
- Digital signature
- Confirmation checkbox
- Re-authentication

Example:

☒ I confirm that I have reviewed the supporting information.

Final Action

Confirm Approval

The source explicitly requires authentication before approval actions and supports MFA and confirmation for sensitive actions.

Screen 5 — Reject / Return / Request Information

Purpose

Manage situations where an approver cannot or should not approve the request.

Decision Options

Reject

Return for Modification

Request Additional Information

Rejection

The approver must select or enter:

Reason Category

- Policy violation
- Insufficient evidence
- Incorrect amount
- Unauthorized request
- Duplicate request
- Risk concern
- Other

Mandatory Comment

The source specifically requires rejection reasons when an approval is denied.

Request Information

Example:

Please attach the original customer complaint and manager authorization.

The request may then enter:

Waiting for Requester

rather than being rejected.

Screen 6 — Requester Modification & Resubmission

Purpose

Allow rejected or returned requests to be corrected and submitted again.

The source specifically requires rejected requests to be modified and resubmitted.

Original Request

Refund Amount: AED 7,500

Rejection Reason: Supporting documentation missing.

Requester Changes

- Add attachment
- Modify business reason
- Correct amount
- Update supporting information
- Add comments

Change Comparison

Original

AED 7,500

Revised

AED 6,800

Resubmission

Save Draft | Resubmit

TICVAI shall maintain the original request and previous decision rather than overwriting them.

Screen 7 — Withdrawal, Cancellation, Expiration & Reopening

Purpose

Manage non-standard approval lifecycle actions.

The matrix explicitly covers draft requests, cancellation, expiration and reopening.

Available Actions

Withdraw

Requester withdraws a pending request.

Example:

Customer no longer requires refund.

Cancel

Authorized administrator cancels the approval process.

Expire

System automatically expires the request after the configured validity period.

Reopen

Authorized user reopens a previously completed request.

Lifecycle

Draft → Submitted → Pending → Approved / Rejected

Alternative:

Pending → Withdrawn

Pending → Cancelled

Pending → Expired

Completed → Reopened

Every transition shall be auditable.

Screen 8 — Segregation of Duties & Four-Eyes Control

Purpose

Prevent inappropriate or conflicting approval actions.

This is a critical governance screen.

The matrix requires the system to prevent users from approving their own requests and supports dual approval for sensitive operations.

Example Warning

Approval Blocked

You cannot approve this request because you initiated the transaction.

Four-Eyes Example**Refund: AED 18,500**

Required:

Approver 1 — Finance Director ✓**Approver 2 — General Manager** Pending**Execution Status****Locked until second approval****Conflict Checks**

TICVAI should detect:

- Requester = approver
- Transaction creator = approver
- Same person occupying two required approval stages
- Delegation creating conflict
- Required independent approver unavailable
- Authority exceeded

Screen 9 — Approved Action Execution & Status**Purpose**Separate **approval of the request** from **execution of the underlying business transaction**.

This distinction is extremely important.

An approval being completed does not automatically mean the underlying transaction executed successfully.

Example**Approval**

Refund AED 7,500

Approval Status

✓ Fully Approved

Execution Status**Processing**

↓

Finance API called

↓

Payment Gateway contacted

↓

Refund completed

Result**Successfully Executed**

Transaction ID: REF-93821

Executed: 14:32

Possible Execution States

Pending | Processing | Successful | Failed | Partially Completed | Retry Required

If execution fails:

Refund approved but payment gateway execution failed.

Actions:

Retry | Investigate | Escalate

This is important for TICVAI because approval governance should never falsely report that a business action happened merely because somebody approved it.

Screen 10 — Decision Record & Immutable Audit View

Purpose

Provide the authoritative record of a completed approval.

Approval Record

Request: APR-10543

Workflow: Refund Approval V5

Final Outcome: Approved

Decision Chain

Stage	Approver	Decision	Time
1	Finance Manager	Approved	10:22
2	Finance Director	Approved	10:41
3	GM	Approved	11:03

Governance Evidence

- Request data
- Original transaction
- Rules applied
- Approval authority
- AI recommendation
- Risk score
- Comments
- Attachments
- Authentication method
- MFA confirmation
- Digital signature, if applicable
- Execution result
- Complete timestamps

Record Status

Immutable

The matrix requires records suitable for regulatory audits, completed records that cannot be modified, digital-signature support and configurable record retention.

It additionally requires tamper detection for unauthorized modification attempts.

Board 4 — Screen Summary

#	Screen
1	Approval Decision Workspace
2	Business Context & Evidence Viewer
3	Approval Timeline & Decision Chain
4	Approve & Sensitive Action Confirmation
5	Reject / Return / Request Information
6	Requester Modification & Resubmission
7	Withdrawal, Cancellation, Expiration & Reopening
8	Segregation of Duties & Four-Eyes Control
9	Approved Action Execution & Status
10	Decision Record & Immutable Audit View

Board 4 Acceptance Criteria

Board 4 shall be considered complete when TICVAI can provide an approver with the complete business context necessary to make a decision; record approval, rejection and supporting comments; require rejection reasons; request additional information; allow authorized modification and resubmission; control withdrawal, cancellation, expiration and reopening; prevent self-approval and other segregation-of-duty violations; enforce dual approval where configured; apply authentication/MFA requirements to sensitive decisions; execute the underlying approved action in a controlled manner; and retain an immutable, auditable decision record.

Important Design Principle

There are actually **three different events** here

1. Decision

Finance Director clicked **Approve**.

↓

2. Governance Completion

All required approval stages have been satisfied.

↓

3. Business Execution

The AED 7,500 refund was actually processed successfully.

That separation will prevent serious operational and financial problems later.

Board 5 should then focus on Delegation, Availability, SLA & Escalation Management — covering temporary delegation, out-of-office routing, SLA policies, reminders, multi-level escalation, breach handling and escalation performance.

Module 11 — Approval Workflows & Governance

Board 5 — Delegation, Availability, SLA & Escalation Management



Board Objective

Board 5 shall provide a centralized configuration and operational environment for managing **approver availability, temporary delegation, substitute approvers, approval SLAs, reminders, automatic rerouting, and multi-level escalation**.

The objective is to ensure that an approval process **never becomes stuck because an approver is absent, unavailable, overloaded, or fails to act within the required time**.

The source explicitly requires approval delegation, temporary delegation with automatic expiration, out-of-office routing, configurable approval SLAs, breach alerts, multi-level escalation, and escalation history/reporting/analytics.

Screen 1 — Delegation & Escalation Command Center

Purpose

Provide administrators and managers with a real-time overview of delegation, SLA and escalation conditions across TICVAI.

KPI Cards

- Active Delegations
- Approvers Unavailable

- Pending Approvals
- SLA At Risk
- SLA Breached
- Escalated Today
- Unassigned Requests
- Critical Escalations

Operational Widgets

Approver Availability

SLA Health

Escalation Volume

Delegations Expiring Soon

Approval Bottlenecks

AI Insights

Finance approval queue is predicted to breach SLA within 47 minutes.

12 requests are currently assigned to an approver who becomes unavailable at 17:00.

Refund approvals at Waterpark have 31% more escalations than the tenant average.

Screen 2 — Delegation Management

Purpose

Allow an authorized approver or administrator to delegate approval authority to another eligible user.

The source specifically requires approvers to be able to delegate their approval authority.

Delegation Setup

Delegator: Ahmed Raza

Delegate To: Sara Khan

Delegation Scope

Choose:

- All approvals
- Specific workflow
- Approval type
- Department
- Venue
- Value range
- Approval authority level

Example:

Delegate Refund Approvals

Up to AED 5,000

Dubai Theme Park only

Controls

- Allow / prohibit onward delegation
- Maximum delegated authority

- Require manager approval
- Notify affected users

Actions:

Save Draft | Activate Delegation | Cancel

Screen 3 — Temporary Delegation & Availability Calendar

Purpose

Manage time-limited delegation caused by annual leave, business travel, training, sickness, or other planned absence.

The matrix requires temporary delegation periods with automatic expiration.

Calendar

Display approver availability across:

Day | Week | Month

Example:

Ahmed Raza

Unavailable:

10–17 September

Delegate:

Sara Khan

Effective:

10 Sep 00:00

→

17 Sep 23:59

Automatic Behavior

At start:

Delegation automatically activated.

At expiration:

Delegation automatically removed and original approval authority restored.

Conflict Detection

Warn when:

- Delegate is also unavailable.
- Delegate lacks required authority.
- Delegation overlaps another delegation.
- Delegation creates SoD conflict.
- Delegation exceeds permitted authority.

Screen 4 — Out-of-Office & Substitute Routing

Purpose

Configure automatic routing when the primary approver is unavailable.

The matrix specifically requires automatic rerouting when approvers are unavailable.

Routing Example

Primary Approver

Finance Manager — Ahmed

↓

Unavailable?

YES

↓

Substitute 1

Deputy Finance Manager — Sara

↓

Unavailable?

YES

↓

Substitute 2

Finance Director — Omar

Routing Policy

Configure:

- Immediate rerouting
- Wait X minutes
- Route to deputy
- Route to manager
- Route to shared queue
- Escalate
- Keep original approver informed

Availability Sources

TICVAI can use:

- Manual availability status
- Delegation calendar
- Out-of-office configuration
- User account status
- Shift/working schedule where integrated

Screen 5 — Approval SLA Policy Configuration

Purpose

Define how quickly different approval types must be processed.

The matrix explicitly requires configurable approval response-time targets.

Example

Workflow: Refund Approval

Approval Stage	Target
Supervisor	15 min
Finance Manager	1 hour
Finance Director	2 hours
GM	4 hours

SLA Configuration

Configure:

- SLA duration
- Business hours / calendar hours
- Working days
- Weekends
- Public holidays
- Venue operating hours
- Priority-based SLA
- Risk-based SLA

Example

Normal Refund:

2 Hours

High Risk:

30 Minutes

Critical:

15 Minutes

Screen 6 — Reminder & Breach Notification Rules

Purpose

Configure proactive notifications **before and after SLA breaches**.

The source requires reminders for pending approvals and alerts when SLA thresholds are exceeded.

Example Timeline

Approval Assigned

0 min

↓

50% SLA

Reminder to approver

↓

75% SLA

Urgent reminder

↓

90% SLA

Approver + manager warning

↓

100% SLA

SLA Breach

↓

Automatic Escalation

Notification Recipients

- Approver
- Requester
- Manager
- Department head
- Workflow owner
- Escalation approver

Notification Channels

- In-app
- Email
- Mobile push
- Integrated enterprise channels where configured

Screen 7 — Escalation Policy Builder

Purpose

Define what TICVAI should do when an approval remains unresolved or meets another escalation condition.

Trigger Types

SLA Breach

High Risk

High Value

Approver Unavailable

Repeated Rejection

Manual Escalation

AI Recommendation

Example

Refund > AED 10,000

Finance Manager

↓

No action after 1 hour

↓

Finance Director

↓

No action after 2 hours

↓

CFO

↓

No action after 4 hours

↓

General Manager

The requirements explicitly support escalation through multiple organizational levels.

Escalation Actions

- Add approver
- Replace approver
- Increase priority
- Notify manager
- Route to shared queue
- Require additional approval
- Trigger critical alert

Screen 8 — Live Escalation Operations Center

Purpose

Allow managers to monitor and intervene in currently escalated approvals.

Queue

Request	Current Level	Waiting	Reason	Next Action
APR-10821	L2	1h 42m	SLA	CFO
APR-10826	L3	3h 12m	High Risk	GM
APR-10831	L1	28m	Unavailability	Director

Request Timeline**Submitted**

↓

Finance Manager

↓

SLA Breached

↓

Finance Director

↓

Current**Manager Actions****Reassign****Escalate Now****Extend SLA****Add Approver****Resolve**

Any manual intervention must be recorded in the audit trail.

Screen 9 — SLA & Escalation Performance Analytics

Purpose

Provide management with analytics about where approval delays and escalations occur.

The source explicitly requires escalation reporting and escalation trend analytics.

KPIs

SLA Compliance

94.2%

Average Response

37 min

Escalation Rate

8.4%

Breach Rate

5.8%

Analytics

SLA by Department

Finance — 91%

Operations — 97%

Commercial — 88%

IT — 95%

Escalations by Reason

- SLA breach
- Approver unavailable
- High risk
- High value
- Manual escalation

Bottleneck View

Identify:

- Slowest approvers

- Slowest stages
- Slowest workflows
- High-escalation venues
- High-escalation departments
- Peak approval periods

Screen 10 — AI SLA & Escalation Advisor

Purpose

Use AI to predict approval bottlenecks and recommend preventive actions before SLAs are breached.

The matrix specifically requires AI escalation recommendations.

Example AI Alert

Predicted SLA Risk — HIGH

Finance currently has 47 pending refund approvals.

Based on current processing speed, approximately **16 requests are likely to breach SLA within the next 60 minutes.**

AI Recommendation

Temporarily redistribute 12 approvals from Finance Manager Ahmed to Deputy Finance Manager Sara.

Expected Impact

Predicted breaches

16 → 4

Average wait time

48 min → **27 min**

Another Example

Recurring Bottleneck Detected

Commercial Director approval accounts for 38% of discount-override SLA breaches.

Recommendation

Consider increasing Commercial Manager authority from 15% to 20%.

Actions

Simulate Recommendation

Create Draft Change

Send for Approval

Dismiss

Governance Principle

AI should not automatically change approval authority or escalation rules unless the client explicitly enables such automation.

The safer default is:

AI Detects → AI Recommends → Authorized User Reviews → Governance Approval → Change Activated

Board 5 — Screen Summary

#	Screen
1	Delegation & Escalation Command Center
2	Delegation Management
3	Temporary Delegation & Availability Calendar
4	Out-of-Office & Substitute Routing
5	Approval SLA Policy Configuration
6	Reminder & Breach Notification Rules
7	Escalation Policy Builder
8	Live Escalation Operations Center
9	SLA & Escalation Performance Analytics
10	AI SLA & Escalation Advisor

Board 5 Acceptance Criteria

Board 5 shall be considered complete when TICVAI can allow authorized users to delegate approval authority within controlled limits; activate and expire temporary delegation automatically; identify unavailable approvers and reroute requests; configure SLA targets by workflow and approval stage; issue reminders and breach alerts; configure multi-level escalation paths; provide real-time visibility of escalated requests; maintain escalation history; analyze SLA and escalation performance; and use AI to predict bottlenecks and recommend corrective actions.

One Important Architecture Rule

I recommend that development team implement a **“No Dead-End Approval” principle**.

Every active approval workflow should always have a valid fallback route:

Primary Approver

→ Delegate

→ Substitute

→ Manager / Next Authority

→ Shared Approval Center

→ Final Escalation Authority

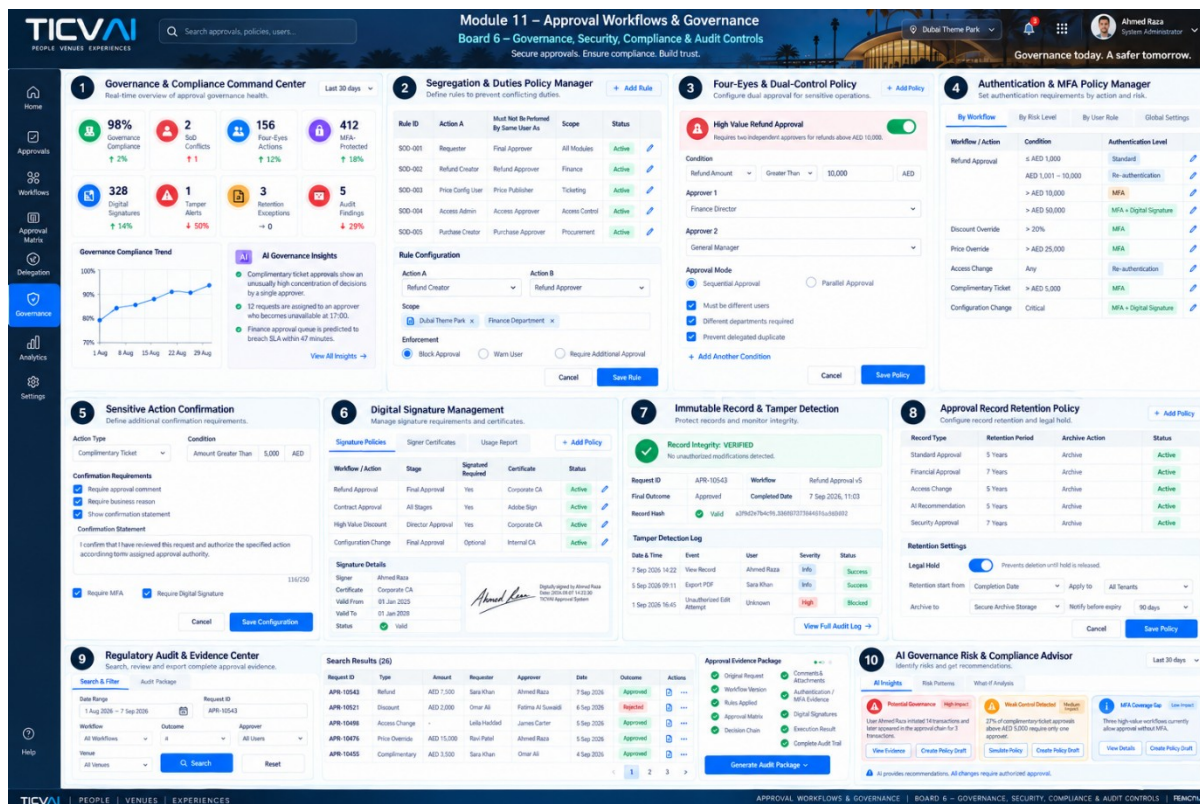
If TICVAI detects that a workflow could reach a point where **no eligible approver exists**, the configuration should be flagged before publication.

That will be particularly important for large clients operating multiple venues, departments, shifts and approval hierarchies.

Board 6 should next cover Governance, Security, Compliance & Audit Controls — segregation of duties, Four-Eyes controls, immutable records, regulatory audit, digital signatures, authentication/MFA, tamper detection and record-retention policies.

Module 11 — Approval Workflows & Governance

Board 6 — Governance, Security, Compliance & Audit Controls



Board Objective

Board 6 shall provide the centralized control environment for protecting the **integrity, security, accountability, and regulatory defensibility** of TICVAI approval processes.

The board governs sensitive approvals through **Segregation of Duties (SoD), Four-Eyes controls, authentication/MFA, digital signatures, immutable records, tamper detection, retention policies, and regulatory audit support.**

The source explicitly requires self-approval prevention, dual approvals for sensitive operations, regulatory audit support, immutable records, digital signatures, configurable retention, authentication, MFA, sensitive-action confirmation and tamper detection.

Screen 1 — Governance & Compliance Command Center

Purpose

Provide administrators, auditors and security teams with an overview of approval governance health across the organization.

KPI Cards

- Governance Compliance %
- SoD Conflicts
- Four-Eyes Protected Actions
- MFA-Protected Approvals
- Digital Signatures
- Tamper Alerts
- Retention Exceptions
- Audit Findings

Governance Health

Display by:

- Tenant
- Venue
- Department
- Workflow
- Approval type
- Risk classification

Example Alerts

2 potential Segregation-of-Duties conflicts detected.

4 sensitive workflows do not currently require MFA.

1 approval record retention policy expires within 30 days.

AI Governance Insight

Complimentary-ticket approvals show an unusually high concentration of decisions by a single approver.

Screen 2 — Segregation of Duties Policy Manager

Purpose

Define combinations of actions that the same person must not be allowed to perform.

The matrix explicitly states that users shall be prevented from approving their own requests.

Example Rules

Rule SOD-001

Requester

≠

Final Approver

Rule SOD-002

Refund Creator

≠

Refund Approver

Rule SOD-003

Price Configuration User

≠

Price Publication Approver

Rule SOD-004

Access Administrator

≠

Access Permission Approver

Rule Configuration

Administrators configure:

Action A

Must Not Be Performed By Same User As

Action B

Scope

- Tenant
- Venue
- Department
- Role
- Workflow
- Transaction type

Enforcement

Block | Warn | Require Additional Approval

Screen 3 — Four-Eyes & Dual-Control Policy

Purpose

Configure sensitive actions requiring approval from **at least two independent authorized persons**.

The source explicitly requires dual approval capability for sensitive operations.

Example

Refund > AED 10,000

Approver 1

Finance Director

-

Approver 2

General Manager

Independence Rules

Administrators should configure:

- Two different users required
- Different roles required
- Different departments if required
- No delegated duplicate identity
- Minimum authority level

- Sequential or parallel approval
- Both approvals mandatory

Execution Control

Until both approvals are complete:

BUSINESS ACTION LOCKED

Screen 4 — Authentication & MFA Policy Manager

Purpose

Define the authentication strength required before an approver can execute sensitive approval decisions.

The source requires authentication before approval actions and supports MFA for sensitive approvals.

Authentication Levels

Level 1 — Standard Authentication

Normal session authentication.

Level 2 — Re-Authentication

Require password / enterprise authentication again.

Level 3 — MFA

Require second factor.

Level 4 — High Assurance

MFA + explicit confirmation and, where configured, digital signature.

Example Policy

Refund:

≤ AED 1,000 → Standard

AED 1,001–10,000 → Re-authentication

> AED 10,000 → MFA

> AED 50,000 → MFA + Digital Signature

Conditions

Can also depend on:

- Risk score
- Workflow
- approval type
- user role
- venue
- device/session context

Screen 5 — Sensitive Action Confirmation

Purpose

Configure the final confirmation requirements before a sensitive approval becomes binding.

The source specifically requires confirmation before execution of sensitive approvals.

Policy Example**Action: Complimentary Ticket > AED 5,000**

Require:

- ☒ Approval comment
- ☒ Business reason
- ☒ Confirmation statement
- ☒ MFA
- ☒ Digital signature

Confirmation Statement

I confirm that I have reviewed this request and authorize the specified action according to my assigned approval authority.

Configuration

Administrators can define different confirmation requirements by:

- Action
- Amount
- Risk
- department
- venue
- tenant

Screen 6 — Digital Signature Management

Purpose

Manage workflows where approval decisions require a digital signature.

The matrix explicitly requires digital-signature support for sensitive approvals.

Signature Policy

Configure:

- Workflow requiring signature
- Approval stage
- Signer role
- Signature requirement
- Certificate information
- Timestamp
- Signature validation status

Example**Capital Refund > AED 50,000**

Finance Director ✓

CFO ✓

Digital Signature:

Required

Signature Record

Store:

- Signer
- Date/time
- Approval request
- Decision
- Signature reference
- Verification status

Status

Verified | Pending | Failed | Revoked

Screen 7 — Immutable Approval Record & Tamper Detection

Purpose

Protect completed approval records from unauthorized modification.

The matrix requires completed approval records to be immutable and requires detection of unauthorized modification attempts.

Record Example

APR-10543

Status: **Completed**

Immutable Record

Integrity Information

- Record ID
- Workflow version
- Request hash/reference
- Completion timestamp
- Decision chain
- Authentication evidence
- Signature evidence
- Execution result
- Integrity status

Tamper Monitoring

Display:

Record Integrity: VERIFIED ✓

If an unauthorized modification is attempted:

Potential Tamper Event Detected

Display:

- Record affected
- User/service
- Date/time

- attempted action
- source
- severity
- investigation status

Actions

Investigate | Lock Access | Export Evidence | Escalate

Screen 8 — Approval Record Retention Policy

Purpose

Define how long approval and governance records shall be retained.

The matrix explicitly requires configurable approval record-retention policies.

Example

Record Type	Retention
Standard Approval	5 Years
Financial Approval	7 Years
Access Change	5 Years
AI Recommendation Approval	5 Years
Security Approval	7 Years

Configuration

- Retention duration
- Tenant
- Country / jurisdiction
- Record category
- Workflow
- Effective date
- archival behavior
- disposal behavior

Legal / Compliance Hold

Administrators should also be able to place a record under:

Legal Hold

When active, scheduled deletion/disposal is prevented until the hold is released.

This legal-hold capability is my recommended enhancement around the matrix's retention requirement.

Screen 9 — Regulatory Audit & Evidence Center

Purpose

Provide internal and external auditors with controlled access to complete approval evidence.

The source specifically requires approval records suitable for regulatory audits.

Audit Search

Search by:

- Request
- Transaction
- Customer
- Approver
- Requester
- Workflow
- Venue
- Department
- Date
- approval outcome
- risk
- value

Audit Evidence Package

For an approval, TICVAI should compile:

Original Request

-

Workflow Version

-

Rules Applied

-

Approval Matrix

-

Decision Chain

-

Comments

-

Authentication / MFA Evidence

-

Digital Signatures

-

Execution Result

-

Complete Audit Trail

Export

Generate Audit Package

Possible controlled formats:

PDF | CSV/Data Export | Evidence Package

Every export itself should also be logged.

Screen 10 — Governance Risk & AI Compliance Advisor

Purpose

Use AI to identify unusual governance patterns, potential policy weaknesses and compliance risks.

This screen should **assist compliance teams rather than automatically change governance policies.**

Example AI Finding 1

Potential Governance Risk

User Ahmed initiated 14 transactions and later appeared in the approval chain for 3 related transactions.

Recommendation

Review SoD policy configuration.

Example AI Finding 2

Weak Control Detected

27% of complimentary-ticket approvals above AED 5,000 require only one approver.

Recommendation

Apply Four-Eyes control above AED 5,000.

Example AI Finding 3

MFA Coverage Gap

Three high-value workflows currently allow approval without MFA.

Recommended Action

Apply MFA to:

- Refund > AED 10,000
- Complimentary > AED 5,000
- Price Override > 25%

Actions

Review Finding

View Evidence

Simulate Policy

Create Draft Policy

Send for Approval

Dismiss

AI should never silently modify governance rules.

Board 6 — Screen Summary

#	Screen
1	Governance & Compliance Command Center
2	Segregation of Duties Policy Manager
3	Four-Eyes & Dual-Control Policy
4	Authentication & MFA Policy Manager
5	Sensitive Action Confirmation
6	Digital Signature Management
7	Immutable Approval Record & Tamper Detection
8	Approval Record Retention Policy
9	Regulatory Audit & Evidence Center
10	Governance Risk & AI Compliance Advisor

Board 6 Acceptance Criteria

Board 6 shall be considered complete when TICVAI can:

- Prevent users from approving their own requests and enforce configurable SoD rules.
- Require independent dual approval for sensitive actions.
- Apply different authentication and MFA requirements according to action, amount and risk.
- Require explicit confirmation before sensitive decisions.
- Support digital signatures where configured.
- Protect completed approval records against modification.
- Detect and record potential tampering attempts.
- Apply configurable record-retention policies.
- Produce complete evidence packages for audit purposes.
- Use AI to identify potential governance weaknesses while keeping policy changes under human approval.

Important Architecture Principle

I recommend development team treat this board as a **cross-platform Governance Control Layer**, rather than building security separately inside Refund, Pricing, Access Control, Membership, etc.

For example:

Refund Request → Approval Engine → Governance Controls → Decision → Execution

AI Price Recommendation → Approval Engine → Governance Controls → Decision → Publication

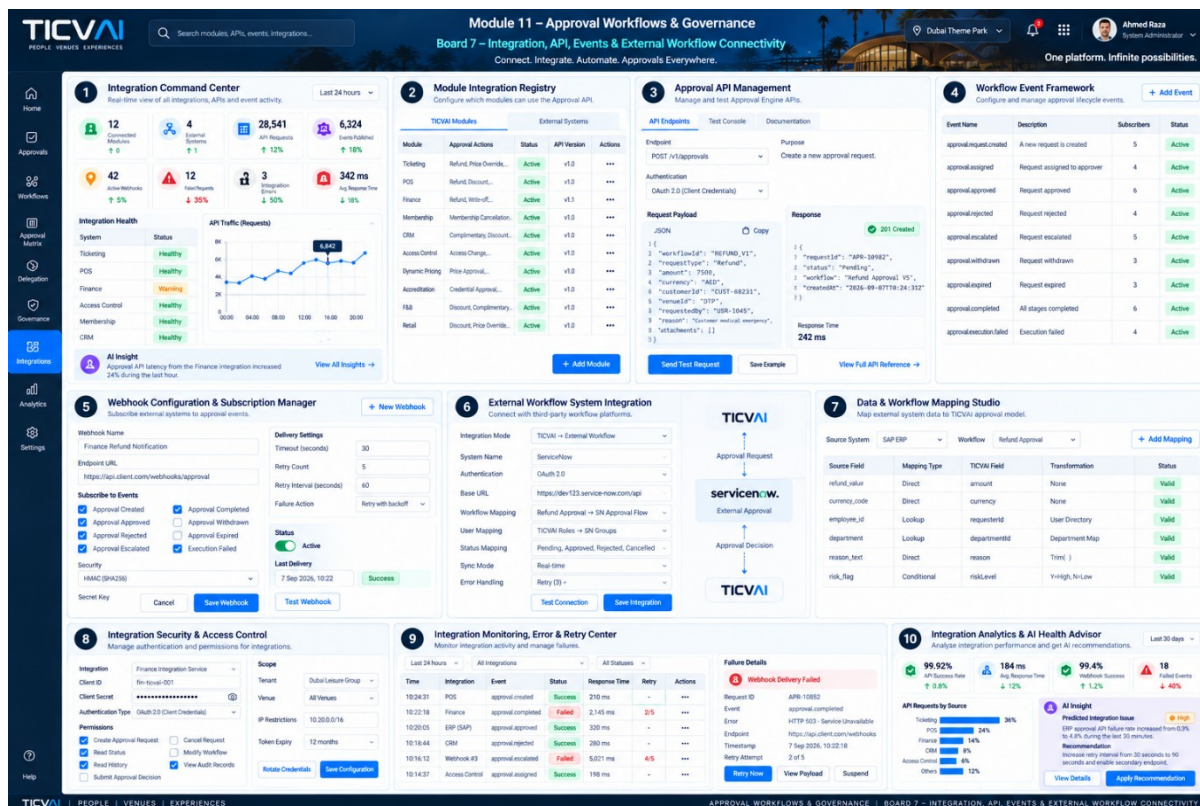
Access Permission Change → Approval Engine → Governance Controls → Decision → Access System

This gives TICVAI a single auditable control framework across the platform.

Board 7 should next cover Integration, API, Events & External Workflow Connectivity — API-based approval processing, workflow events, webhooks, third-party workflow systems and integration monitoring, which are explicitly required in 11.1.63–11.1.66.

Module 11 — Approval Workflows & Governance

Board 7 — Integration, API, Events & External Workflow Connectivity



Board Objective

Board 7 shall provide the technical administration environment for connecting the **TICVAI Approval & Governance Engine** with all **TICVAI modules** and **authorized external systems**.

This board is important because the Approval Engine should operate as a **central platform service**. Ticketing, POS, Finance, Access Control, CRM, Membership, Dynamic Pricing, AI and other modules should call the same governance engine rather than implementing their own approval mechanisms.

The source specifically requires API-based approval processing, workflow event publication, third-party workflow integration and approval webhooks.

Screen 1 — Approval Integration Command Center

Purpose

Provide administrators and technical teams with a centralized overview of all systems connected to the Approval Engine.

KPI Cards

- Connected TICVAI Modules
- External Integrations
- API Requests Today
- Workflow Events
- Active Webhooks
- Failed Requests
- Integration Errors
- Average Response Time

Connected Services

Example:

System	Integration	Status	Activity
Ticketing	Internal API	Connected	1,284
POS	Internal API	Connected	842
Finance	Internal API	Connected	326
Access Control	Event/API	Connected	74
Dynamic Pricing	API	Connected	48
External ERP	REST API	Connected	126

Health Status

Healthy | Warning | Degraded | Failed

AI Insight

Approval API latency from the Finance integration increased 24% during the last hour.

Screen 2 — Module Integration Registry

Purpose

Define which TICVAI modules are allowed to initiate and consume approval workflows.

Example

Module: Ticketing

Supported approval actions:

- ☒ Refund
- ☒ Price Override
- ☒ Complimentary Ticket

☒ Ticket Configuration Change

☒ Product Publication

Other Modules

- Finance
- POS
- Membership
- CRM
- Promotions
- Access Control
- Accreditation
- Dynamic Pricing
- Resource Management
- F&B
- Retail
- AI Engine

Configuration

For each module:

- Module ID
- Allowed workflow types
- Request permissions
- Response permissions
- Tenant scope
- Venue scope
- Authentication method
- API version
- **Status**

This screen establishes the Approval Engine as a reusable TICVAI platform service.

Screen 3 — Approval API Management

Purpose

Configure and monitor APIs used to create, retrieve and process approval requests.

The matrix explicitly requires approval workflows to be exposed through APIs.

Core API Operations

Conceptually:

Create Approval Request

Get Approval Status

Get Approval Details

Submit Decision

Withdraw Request

Cancel Request**Get Approval History****API Configuration**

- Endpoint
- API version
- Authentication
- tenant context
- request schema
- response schema
- timeout
- retry policy
- rate limit

API Testing

Provide:

Request Payload

→ **Send Test**

→ **Response**

Example:**201 — Approval Request Created**

Request:

APR-10982

Workflow:

Refund Approval V5

Status:

Pending Approval

Screen 4 — Workflow Event Framework

Purpose

Configure the business events published by the Approval Engine.

The matrix requires workflow events to be published through the platform event framework.

Event Types**Examples:**

approval.request.created

approval.assigned

approval.approved

approval.rejected

approval.escalated

approval.withdrawn

approval.expired

approval.completed**approval.execution.failed****Event Configuration**

For each event:

- Event name
- source
- tenant
- workflow
- payload
- subscribers
- timestamp
- correlation ID
- delivery status

Example**approval.completed**

↓

Finance

↓

POS

↓

CRM

↓

Reporting

Different systems can therefore react automatically to the same approval event.

Screen 5 — Webhook Configuration & Subscription Manager

Purpose

Allow external or internal systems to subscribe to approval events.

The source explicitly requires webhook notifications for approval events.

Create Webhook**Webhook Name**

Finance Refund Notification

Destination

Configured HTTPS endpoint

Subscribe To

- ☒ Approval Created
- ☒ Approved
- ☒ Rejected
- ☒ Escalated

☒ Completed

Security

- Authentication secret
- Signature validation
- IP restrictions where applicable
- certificate/security configuration

Delivery Configuration

- Timeout
- Retry count
- Retry interval
- failure action
- disable threshold

Status

Active | Suspended | Failed

Screen 6 — External Workflow System Integration

Purpose

Allow TICVAI to integrate with third-party workflow/governance platforms when a client already operates an enterprise approval environment.

The matrix specifically requires integration with third-party workflow systems.

Integration Modes

TICVAI → External Workflow

TICVAI creates the business request.

↓

External platform performs approval.

↓

Decision returned to TICVAI.

↓

TICVAI executes the business action.

External System → TICVAI

External system initiates a request.

↓

TICVAI Governance Engine processes approval.

↓

Outcome returned externally.

Configuration

- System name
- integration direction

- authentication
- workflow mapping
- user/role mapping
- status mapping
- error handling
- synchronization mode

Important

TICVAI must retain sufficient audit information even when the approval decision occurs externally.

Screen 7 — Data & Workflow Mapping Studio

Purpose

Map external system data to TICVAI's approval data model without requiring custom development for every integration.

Example

External ERP:

refund_value

→

TICVAI:

Request.Amount

External:

employee_department

→

TICVAI:

Requester.Department

External:

risk_level

→

TICVAI:

Risk.Score

Mapping Areas

- Request fields
- users
- roles
- departments
- venues
- workflow types
- statuses
- decisions
- currencies

- timestamps
- reason codes

Transformation

Allow:

Direct Mapping**Lookup****Default Value****Conditional Mapping****Value Transformation****Validation**

- Required field missing
- Unmapped status
- Unknown approver role
- Mapping valid

Screen 8 — Integration Security & Access Control

Purpose

Govern which applications and external systems can access the Approval Engine.

Application Identity

Example:

Finance Integration Service**Permissions**

- ☒ Create Approval Request
- ☒ Read Status
- ☒ Read History
- ☐ Submit Approval Decision
- ☐ Modify Workflow

Scope

Tenant: Dubai Leisure Group

Venue: All Venues

Module: Finance

Security Controls

- API credentials
- OAuth/token configuration where applicable
- certificate authentication
- secret rotation
- IP restrictions
- permission scopes
- expiration

- tenant restrictions

Critical Principle

An integration authorized for **Tenant A** must never access **Tenant B's** approvals.

This supports the source requirement for tenant isolation.

Screen 9 — Integration Monitoring, Error & Retry Center

Purpose

Provide technical teams with real-time monitoring of integration failures.

Operational Queue

Time	Integration	Event	Status	Retry
14:32	Finance	approval.completed	Success	—
14:31	ERP	approval.created	Failed	2/5
14:29	CRM	approval.approved	Success	—
14:26	Webhook #12	escalation	Failed	4/5

Failure Detail

Example:

Webhook Delivery Failed

Request:

APR-10852

Event:

approval.completed

Error:

HTTP 503

Actions

Retry Now

View Payload

View Response

Suspend Integration

Escalate

Retry Policy

TICVAI should support controlled retries and ensure duplicate delivery does not result in duplicate business execution.

I strongly recommend **idempotency controls** here, particularly for refunds and financial operations.

Screen 10 — Integration Analytics & AI Health Advisor

Purpose

Provide technical and business teams with analytics on Approval Engine integration performance.

KPIs

API Success Rate

99.92%

Average Response

184 ms

Webhook Success

99.4%

Failed Events

18

Retry Success

92%

Analytics

- API traffic by module
- approval requests by source
- webhook volume
- event volume
- error trends
- latency
- external system availability

AI Insight Example

Integration Anomaly Detected

ERP approval API failure rate increased from 0.3% to 4.8% during the last 30 minutes.

AI Recommendation

Temporarily increase retry interval from 30 seconds to 90 seconds.

Another example:

Duplicate Request Pattern Detected

POS integration submitted 14 duplicate approval requests during the last hour.

Actions

View Evidence

Simulate Recommendation

Create Configuration Change

Send for Approval

Board 7 — Screen Summary

#	Screen
1	Approval Integration Command Center
2	Module Integration Registry
3	Approval API Management
4	Workflow Event Framework
5	Webhook Configuration & Subscription Manager
6	External Workflow System Integration
7	Data & Workflow Mapping Studio
8	Integration Security & Access Control
9	Integration Monitoring, Error & Retry Center
10	Integration Analytics & AI Health Advisor

Board 7 Acceptance Criteria

Board 7 shall be considered complete when TICVAI can expose the Approval Engine through governed APIs; allow TICVAI modules to create and consume approval requests; publish approval lifecycle events; configure webhook subscriptions; integrate with authorized third-party workflow systems; map external data to the TICVAI approval model; restrict integrations by permissions, tenant and scope; monitor API/event/webhook failures; safely retry failed transactions; and provide analytics and AI-assisted integration-health monitoring.

Important Architecture Recommendation

TICVAI Business Modules

Ticketing · POS · Finance · CRM · Access Control · Membership · Dynamic Pricing · Accreditation · F&B · Retail

↓

Central Approval & Governance API

↓

Workflow Engine → Approval Matrix → Governance Controls → SLA/Escalation → Decision

↓

Event Layer

↓

Source Module + Reporting + External Systems

This means a future TICVAI module can simply integrate with the **Approval API** instead of development team rebuilding approval functionality every time.

It also creates an important rule: **the source business module owns the business transaction; the Approval Engine owns the governance decision.** For example, the Approval Engine authorizes a refund, but the Finance/Payment layer actually executes the refund.

Board 8 should be our final board: Approval Analytics, KPI Intelligence & AI Governance Insights, covering the matrix requirements for KPI dashboards, processing-time analytics, bottleneck analysis, volume/trend analysis, and AI approval/risk/priority/escalation recommendations.

Module 11 — Approval Workflows & Governance

Board 8 — Approval Analytics, KPI Intelligence & AI Governance Insights



Board Objective

Board 8 shall provide the **executive analytics and AI intelligence layer** for the entire Approval Workflows & Governance module.

While Boards 1–7 manage configuration, routing, decisions, governance, escalation, and integrations, Board 8 answers:

“How well is our approval organization performing, where are the bottlenecks and risks, and what should management change?”

The matrix explicitly requires KPI dashboards, processing-time analytics, bottleneck analysis, volume analysis, and trend analysis. It also requires AI-generated approval recommendations, risk assessments, priority scoring, and escalation recommendations.

I recommend the following **10 screens**.

Screen 1 — Approval Executive KPI Dashboard

Purpose

Provide executives and senior management with a high-level view of approval performance across TICVAI.

KPI Cards

Total Approval Requests

12,842

Approval Rate

91.4%

Rejection Rate

5.8%

Escalation Rate

7.2%

SLA Compliance

94.6%

Average Approval Time

37 min

High-Risk Requests

186

AI Recommendations Accepted

73%

Filters

- Tenant
- Venue
- Department
- Module
- Workflow
- Request type
- Date range
- Risk
- Priority

Executive AI Summary

Approval performance improved 8% this month, but Commercial approvals remain the primary organizational bottleneck.

The matrix explicitly requires approval KPI dashboards.

Screen 2 — Approval Volume & Outcome Analytics

Purpose

Analyze how many approval requests are being generated and what happens to them.

Volume Analysis

Display:

Submitted**Approved****Rejected****Withdrawn****Cancelled****Expired****Escalated****Charts****Approval Volume by Module**

Ticketing — 38%

Finance — 24%

Promotions — 14%

Membership — 9%

Access Control — 7%

AI Engine — 5%

Other — 3%

Additional Analysis

- Volume by venue
- Volume by department
- Volume by workflow
- Volume by approver
- Volume by hour/day/week/month
- Outcome distribution

The matrix specifically requires approval volume analysis.

Screen 3 — Approval Processing Time Analytics**Purpose**

Measure exactly how long approval processes and individual stages take.

The matrix explicitly requires processing-time analytics.

KPI Cards**Average Approval Time**

37 min

Median

24 min

P95

2h 14m

Fastest Workflow

Complimentary Ticket

Slowest Workflow

Price Override

Processing Time Breakdown

Example:

Request Created

↓

Queue Waiting — **12 min**

↓

Finance Manager — **18 min**

↓

Finance Director — **42 min**

↓

Execution — **3 min**

Analysis

Compare processing time by:

- Workflow
- approval stage
- department
- venue
- approver
- priority
- risk level

This lets management understand whether delays come from **workflow design, workload, or specific approval stages**.

Screen 4 — Bottleneck Analysis & Heatmap

Purpose

Automatically identify where approval processes are becoming slow or congested.

The source explicitly requires approval bottleneck analysis.

Heatmap

Rows:

- Finance
- Commercial
- Operations
- IT
- Security
- Guest Services

Columns:

- Refund
- Discount
- Price Override
- Complimentary
- Access Change
- Configuration

Cells:

Healthy
Watch
Bottleneck
Critical

Top Bottlenecks

Stage	Avg Wait	SLA Breach
Commercial Director	2h 14m	22%
Finance Director	1h 42m	17%
GM Approval	1h 28m	14%

AI Explanation

Commercial Director receives 43% more approval requests than comparable authority levels.

Screen 5 — Approval Trend & Comparative Analytics

Purpose

Identify whether approval performance is improving or deteriorating over time.
The matrix specifically requires approval trend reporting.

Trends

Approval Volume

↑ 12%

Average Approval Time

↓ 8%

SLA Breaches

↑ 3%

Escalations

↓ 6%

Comparison Modes

Today vs Yesterday

This Week vs Last Week

This Month vs Last Month

Venue vs Venue

Department vs Department Workflow vs Workflow

Example

Dubai Theme Park

SLA Compliance: 96%

vs.

Waterpark

SLA Compliance: 87%

This allows management to identify operational differences between venues.

Screen 6 — Approver & Team Performance Analytics

Purpose

Understand workload and operational performance at approver and team level without reducing governance to a simplistic employee ranking.

Metrics

- Assigned requests
- Completed approvals
- Average response time
- SLA compliance
- escalation rate
- workload
- pending queue
- delegation frequency
- approval/rejection ratio

Example

Approver	Queue	Avg Time	SLA	Escalation
Ahmed	28	42m	91%	8%
Sara	17	28m	97%	3%
Omar	34	1h 12m	84%	16%

Important Design Principle

TICVAI should distinguish:

Approver Performance

from

Approver Workload

Someone may have slower response times because the approval matrix is routing an excessive number of requests to that person.

AI should identify this distinction.

Screen 7 — Risk & Governance Analytics

Purpose

Provide management with consolidated visibility of approval risk.

Risk Distribution

Low — 71%

Medium — 20%

High — 7%

Critical — 2%

Risk Analytics

Analyze by:

- Workflow
- venue
- department
- requester
- approver
- transaction type
- value
- AI risk score

Governance Indicators

- SoD conflicts detected
- Four-Eyes interventions
- MFA-protected decisions
- high-value approvals
- unusual rejection patterns
- repeated override requests
- suspicious approval behavior

The source requires AI-generated risk assessment for approval requests.

Screen 8 — AI Approval Intelligence Center

Purpose

Provide one centralized location for AI-generated approval intelligence.

This should be one of the strongest screens in the entire module.

AI Recommendation

APR-10982 — Refund AED 8,500

AI Recommendation

APPROVE

Confidence: **93%**

AI Risk

24 / 100 — LOW

AI Priority

HIGH

Reasoning Summary

- Transaction appears legitimate.
- Refund complies with configured policy.
- Customer refund frequency is normal.
- No duplicate transaction detected.
- Value is within the approver's authority.
- SLA is approaching threshold.

The matrix explicitly requires:

AI Approval Recommendations

AI Risk Assessment

AI Priority Scoring

AI Escalation Recommendations.

Important

AI recommendation must remain distinguishable from the **actual human decision**.

Screen 9 — AI Optimization & What-If Simulator

Purpose

Allow management to evaluate proposed governance changes before implementing them.

AI Finding

Finance Director currently handles 42% of all refund approvals.

Recommendation

Increase Finance Manager authority:

Current

AED 5,000

→

Proposed

AED 7,500

What-If Simulation

Based on historical approval activity:

Director workload

42% → **27%**

Average approval time

47 min → **31 min**

SLA breaches

8.2% → **4.9%**

Estimated risk impact

Low

Alternative Scenarios

Management can test:

- Change threshold
- Add approver
- remove approval stage
- change SLA
- change escalation time
- introduce parallel approval
- modify authority
- require Four-Eyes

Actions

Simulate

Compare Scenarios

Create Draft Change

Send for Approval

This makes AI useful for **designing better governance**, rather than merely displaying statistics.

Screen 10 — AI Governance Executive Advisor

Purpose

Turn all approval analytics into prioritized management recommendations.

Instead of management examining dozens of dashboards, TICVAI AI should answer:

“What requires my attention?”

Critical

Commercial Approval Bottleneck

Commercial Director approval causes 38% of discount-override SLA breaches.

Recommended Action

Increase Commercial Manager authority from 15% to 20%.

Expected Impact

↓ 31% escalations

High Priority

Unusual Refund Pattern

Waterpark has experienced a 22% increase in high-value refund requests during the last seven days.

Recommended Action

Review refund activity and temporarily increase risk monitoring.

Optimization**Approval Stage May Be Unnecessary**

92% of complimentary-ticket requests below AED 500 have been approved without modification during the last six months.

Recommended Action

Evaluate automatic approval below AED 500.

Governance Risk**Control Weakness**

Three high-value configuration workflows do not currently require Four-Eyes approval.

Recommended Action

Introduce independent dual approval.

Management Actions

Investigate

View Evidence

Simulate Impact

Create Draft Change

Send for Approval

Dismiss

Board 8 — Screen Summary

#	Screen
1	Approval Executive KPI Dashboard
2	Approval Volume & Outcome Analytics
3	Approval Processing Time Analytics
4	Bottleneck Analysis & Heatmap
5	Approval Trend & Comparative Analytics
6	Approver & Team Performance Analytics
7	Risk & Governance Analytics
8	AI Approval Intelligence Center
9	AI Optimization & What-If Simulator
10	AI Governance Executive Advisor

Board 8 Acceptance Criteria

Board 8 shall be considered complete when TICVAI can provide management with consolidated approval KPIs; analyze approval volume, outcomes, processing times and trends; identify bottlenecks by workflow, stage,

venue and department; compare performance across organizational dimensions; analyze workload and SLA performance; provide risk and governance analytics; generate AI approval recommendations, risk assessments, priority scores and escalation recommendations; and simulate the operational impact of proposed governance changes before activation.

Any AI-generated configuration change shall remain subject to the appropriate TICVAI approval and governance process.

Module 11 — Complete Board Structure

With Board 8, I consider this module structurally complete:

Board	Scope
1	Approval Command Center
2	Approval Workflow Builder
3	Approval Matrix & Intelligent Routing
4	Approval Execution & Decision Management
5	Delegation, Availability, SLA & Escalation
6	Governance, Security, Compliance & Audit
7	Integration, API, Events & External Connectivity
8	Approval Analytics, KPI Intelligence & AI Governance

That gives us **8 boards / 80 screens**, which aligns very well with the **80 requirements** in the source matrix without forcing a one-requirement-per-screen design. The result is a coherent central **TICVAI Approval & Governance Engine**, rather than 80 disconnected functions.